

Tailwind CSS Practicals with Answers

Utility-First CSS Framework | Web Technology

Utility Classes & Typography · Flexbox & Grid Layout · Forms & Components · Responsive Design & Dark Mode · Tailwind Animations & Custom Config

Practical 1: Utility Classes & Typography

Aim: Apply Tailwind utility classes for text, colors, spacing, and layout.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-gray-100 p-8 font-sans">
  <h1 class="text-4xl font-bold text-blue-700 text-center mb-4">
    Tailwind Typography
  </h1>
  <p class="text-gray-600 text-lg text-center mb-6">
    Using utility classes for styling
  </p>
  <div class="bg-white rounded-xl shadow-md p-6 max-w-md mx-auto">
    <p class="text-sm text-green-600 font-semibold uppercase tracking-wide">
      Category
    </p>
    <h2 class="text-2xl font-bold text-gray-800 mt-1">Card Title</h2>
    <p class="text-gray-500 mt-2 leading-relaxed">
      Tailwind CSS applies styles directly via utility classes,
      no custom CSS needed.
    </p>
    <button class="mt-4 bg-blue-600 hover:bg-blue-700 text-white
      font-semibold py-2 px-6 rounded-lg transition">
      Learn More
    </button>
  </div>
</body>
</html>
```

Explanation:

Tailwind uses utility-first classes: **text-{size}** for font size, **font-{weight}** for weight, **text-{color}-{shade}** for color, **p-{n}/m-{n}** for spacing (1=4px), **rounded-{size}** for border-radius, **shadow-{size}** for box-shadow, **hover:** prefix for hover states, **transition** for smooth animations.

Output: Card renders centered with white background, rounded corners, shadow, and blue button that darkens on hover.

Practical 2: Flexbox & Grid Layout

Aim: Build responsive layouts using Tailwind's flex and grid utility classes.

Code:

```
<!DOCTYPE html>
<html>
<head><script src="https://cdn.tailwindcss.com"></script></head>
<body class="bg-slate-100 p-6">
  <!-- Flexbox Navbar -->
  <nav class="bg-gray-900 flex items-center justify-between px-6 py-3 rounded-xl mb-6">
    <span class="text-yellow-400 font-bold text-lg">Brand</span>
    <div class="flex gap-6">
      <a href="#" class="text-gray-300 hover:text-white text-sm">Home</a>
      <a href="#" class="text-gray-300 hover:text-white text-sm">About</a>
      <a href="#" class="bg-blue-600 text-white px-4 py-1 rounded-lg text-sm">Login</a>
    </div>
  </nav>

  <!-- Grid of Cards -->
  <div class="grid grid-cols-1 md:grid-cols-3 gap-4">
    <div class="bg-white rounded-xl p-5 shadow hover:shadow-lg transition">
      <div class="w-10 h-10 bg-blue-100 rounded-lg flex items-center
        justify-content center mb-3 text-xl">#128187</div>
      <h3 class="font-bold text-gray-800">Web Dev</h3>
      <p class="text-gray-500 text-sm mt-1">Frontend & Backend skills</p>
    </div>
    <div class="bg-white rounded-xl p-5 shadow hover:shadow-lg transition">
```

```


#### Explanation:



flex: display:flex. items-center: align-items:center. justify-between: space-between. gap-{n}: gap. grid: display:grid. grid-cols-3: 3 columns. md:grid-cols-3: 3 cols on medium+ screens (responsive prefix). hover:shadow-lg: shadow on hover.



Output: Navbar with flex layout, logo left, links right. Three cards in a responsive grid (1 col mobile, 3 cols desktop).



## Practical 3: Forms & Components



Aim: Style form elements and common UI components with Tailwind classes.



#### Code:



```

<!DOCTYPE html>
<html>
<head><script src="https://cdn.tailwindcss.com"></script></head>
<body class="bg-gray-50 min-h-screen flex items-center justify-center p-6">
 <div class="bg-white rounded-2xl shadow-lg p-8 w-full max-w-sm">
 <h2 class="text-2xl font-bold text-gray-800 mb-1">Sign In</h2>
 <p class="text-gray-400 text-sm mb-6">Welcome back!</p>

 <!-- Input Field -->
 <div class="mb-4">
 <label class="block text-sm font-medium text-gray-700 mb-1">Email</label>
 <input type="email" placeholder="you@example.com"
 class="w-full border border-gray-300 rounded-lg px-4 py-2 text-sm
 focus:outline-none focus:ring-2 focus:ring-blue-500
 focus:border-transparent transition">
 </div>
 <div class="mb-6">
 <label class="block text-sm font-medium text-gray-700 mb-1">Password</label>
 <input type="password" placeholder="....."
 class="w-full border border-gray-300 rounded-lg px-4 py-2 text-sm
 focus:outline-none focus:ring-2 focus:ring-blue-500 transition">
 </div>

 <!-- Buttons -->
 <button class="w-full bg-blue-600 hover:bg-blue-700 text-white font-semibold
 py-2 rounded-lg transition mb-3">Sign In</button>
 <button class="w-full border border-gray-300 text-gray-700 font-semibold
 py-2 rounded-lg hover:bg-gray-50 transition">
 Create Account
 </button>

 <!-- Badge & Alert -->
 <div class="mt-5 flex gap-2">
 <span class="bg-green-100 text-green-700 text-xs px-2 py-1 rounded-full
 font-semibold">Active
 <span class="bg-yellow-100 text-yellow-700 text-xs px-2 py-1 rounded-full
 font-semibold">Pending
 <span class="bg-red-100 text-red-700 text-xs px-2 py-1 rounded-full
 font-semibold">Inactive
 </div>
 </div>
</body>
</html>

```


```

```
</body>
</html>
```

Explanation:

focus:ring-2: blue glow on focus. **focus:outline-none**: removes default outline. **w-full**: 100% width. **rounded-full**: pill shape for badges. Color sets: **bg-{color}-100** (light bg) + **text-{color}-700** (dark text) = soft badge pattern. **min-h-screen**: full viewport height.

Output: Centered login card with styled inputs (blue ring on focus), solid and outline buttons, and three colored status badges.

Practical 4: Responsive Design & Dark Mode

Aim: Use Tailwind responsive prefixes and dark mode utilities.

Code:

```
<!DOCTYPE html>
<html class="dark">
<head>
  <script src="https://cdn.tailwindcss.com"></script>
  <script>
    tailwind.config = { darkMode: 'class' }
  </script>
</head>
<body class="bg-white dark:bg-gray-900 transition-colors duration-300 p-6">
  <div class="flex justify-between items-center mb-8">
    <h1 class="text-xl font-bold text-gray-900 dark:text-white">My Dashboard</h1>
    <button onclick="document.documentElement.classList.toggle('dark')"
      class="bg-gray-200 dark:bg-gray-700 text-gray-800 dark:text-white
        px-4 py-2 rounded-lg text-sm font-medium transition">
      Toggle Dark
    </button>
  </div>

  <!-- Responsive Stats Grid: 1 col -> 2 col -> 4 col -->
  <div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 mb-8">
    <div class="bg-blue-50 dark:bg-blue-900/30 rounded-xl p-4">
      <p class="text-sm text-blue-600 dark:text-blue-400">Students</p>
      <p class="text-3xl font-bold text-blue-700 dark:text-blue-300">1,248</p>
    </div>
    <div class="bg-green-50 dark:bg-green-900/30 rounded-xl p-4">
      <p class="text-sm text-green-600 dark:text-green-400">Placed</p>
      <p class="text-3xl font-bold text-green-700 dark:text-green-300">847</p>
    </div>
    <div class="bg-purple-50 dark:bg-purple-900/30 rounded-xl p-4">
      <p class="text-sm text-purple-600 dark:text-purple-400">Companies</p>
      <p class="text-3xl font-bold text-purple-700 dark:text-purple-300">63</p>
    </div>
    <div class="bg-orange-50 dark:bg-orange-900/30 rounded-xl p-4">
      <p class="text-sm text-orange-600 dark:text-orange-400">Offers</p>
      <p class="text-3xl font-bold text-orange-700 dark:text-orange-300">924</p>
    </div>
  </div>
</body>
</html>
```

Explanation:

Responsive prefixes: **sm**: (640px+), **md**: (768px+), **lg**: (1024px+), **xl**: (1280px+). **dark**: prefix applies styles when dark mode is active. **darkMode: 'class'** enables class-based dark mode toggle. **duration-300** smooths color transitions.

Output: Stats cards go from 1-column (mobile) to 4-column (desktop). Toggle button switches entire page between light and dark theme instantly.

Practical 5: Tailwind Animations & Custom Config

Aim: Use built-in animations and customize Tailwind via config.

Code:

```
<!DOCTYPE html>
<html>
<head>
```

```

<script src="https://cdn.tailwindcss.com"></script>
<script>
  tailwind.config = {
    theme: {
      extend: {
        colors: { brand: '#7c3aed' },
        animation: { 'spin-slow': 'spin 3s linear infinite' },
        borderRadius: { '4xl': '2rem' }
      }
    }
  }
</script>
</head>
<body class="bg-slate-100 p-8 text-center">
  <!-- Built-in animations -->
  <div class="flex justify-center gap-8 mb-8">
    <div class="w-12 h-12 bg-blue-500 rounded-full animate-bounce"></div>
    <div class="w-12 h-12 bg-purple-500 rounded-lg animate-pulse"></div>
    <div class="w-12 h-12 border-4 border-gray-300 border-t-blue-600
      rounded-full animate-spin"></div>
    <div class="w-12 h-12 bg-green-500 rounded-full animate-ping
      opacity-75"></div>
  </div>

  <!-- Custom brand color -->
  <button class="bg-brand text-white px-8 py-3 rounded-4xl font-bold
    hover:opacity-90 transition mb-4 block mx-auto">
    Custom Brand Button
  </button>

  <!-- Gradient text -->
  <h2 class="text-4xl font-black bg-gradient-to-r from-blue-600
    via-purple-500 to-pink-500 bg-clip-text text-transparent">
    Gradient Text Effect
  </h2>
</body>
</html>

```

Explanation:

Built-in animations: **animate-bounce**, **animate-pulse**, **animate-spin**, **animate-ping**. **tailwind.config** extends defaults — custom colors, border-radius, animations. **bg-gradient-to-r from-{c} to-{c}** + **bg-clip-text text-transparent** creates gradient text. Extended values are available as regular utility classes.

Output: Four animated elements (bounce, pulse, spin, ping). Custom brand-colored button with 4xl radius. Gradient text from blue to pink.

-- End of Practicals --